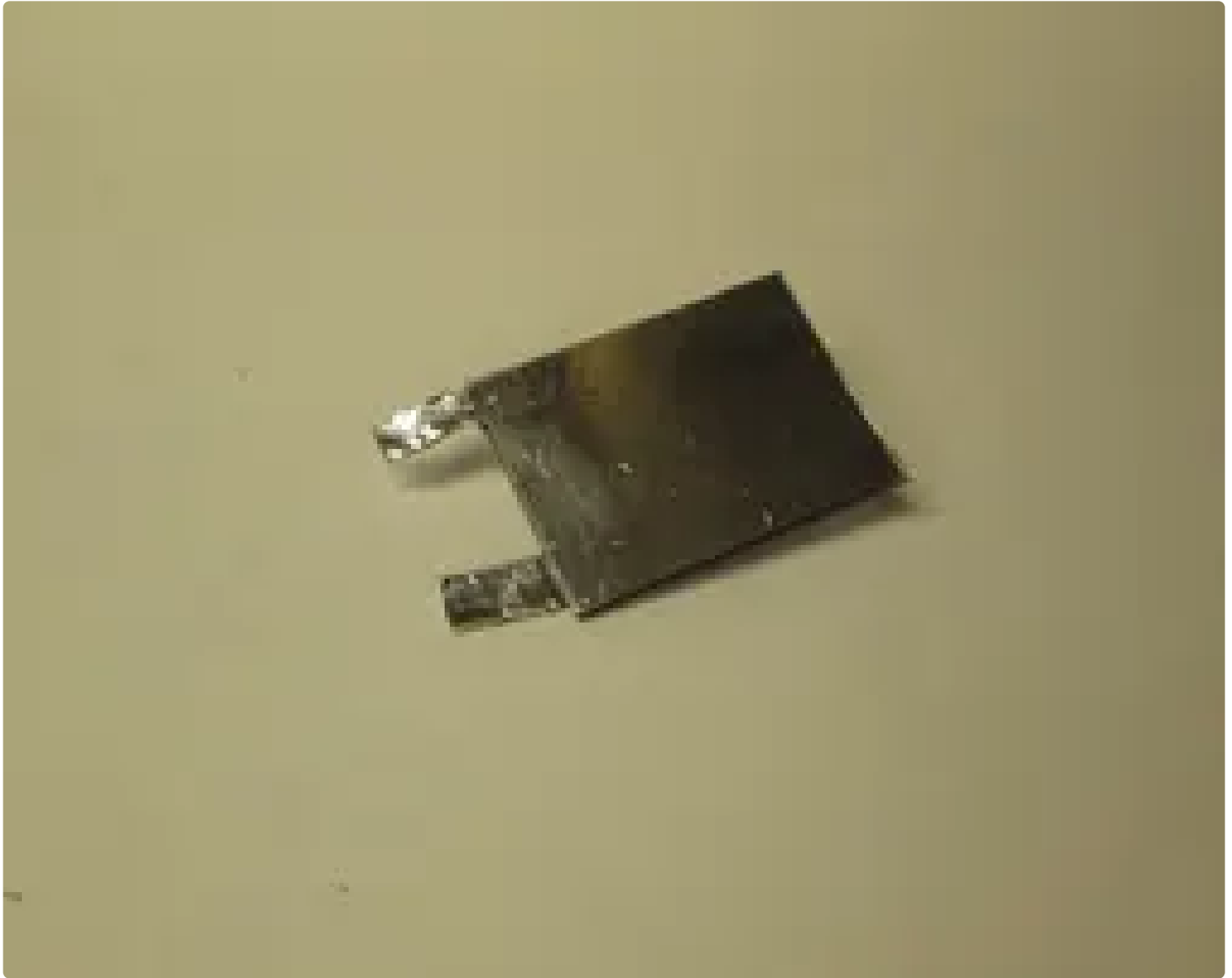


Make a Capacitor With Stuff You Already Have (how It Works+calculations)

By assemblyrequired in Circuits > Electronics  31,835  58  15

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Capacitors are in electronics all around us. As a result, it is important to understand how they work, especially the simplest: the parallel plate capacitor. In this Instructable, I will be showing you how to make your own, and I will also show you

how they work, along with the calculations.

Included in this instructible:

Step 1- How a capacitor works

Step 2- Calculations for a parallel plate capacitor

Step 3- Making a basic parallel plate capacitor

Step 4- Applying calculations from Step 2 onto the capacitor built

All you need is the following:

- Something to act as a plate (any flat conductor) x2
- Something to act as a dielectric (any flat insulator)

We will go over what each of these are in a moment.

- Adhesive, to keep the two together

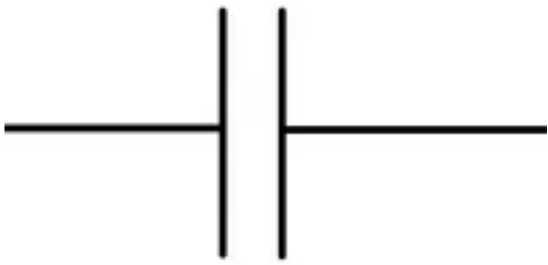
For tools, it would be helpful to have...

- A capacitance meter
- A resistance meter

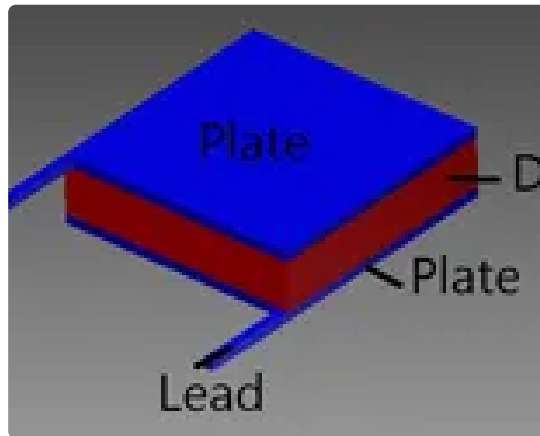
(If you think you don't have one, don't worry- your average multimeter may have one built in. Look for an option to measure in farads, nanofarads, picofarads, etc. for capacitance, and ohms Ω for resistance)

These tools are optional but recommended, as they will only be used to test the capacitor. However, you will definitely need something to cut whatever you choose for the plate and dielectric.

Step 1: What Is a Capacitor?



the symbol for a capacitor
in an engineering schematic



The first image is what a capacitor would look like in an engineering schematic. The second is a screenshot of a capacitor I made in Autocad, showing the parts of a capacitor.

A capacitor is similar to a battery in that it releases electricity. However, where a battery uses chemical reactions to send electrons down a wire, a capacitor takes electricity that is already there and stores it for release. The amount that is released is determined by a number of factors, all of which stem from the main pieces of a capacitor. The three main parts are the dielectric and the two metal plates. These two metal plates are connected to the circuit via the leads.

The dielectric can be any material, as long as it is not a good conductor. Even though this is the case, only some materials are used, as the dielectric constant may not be practical, the electrons used may be able to breakdown the material if a high voltage, and more. (More on the dielectric constant in the calculations page.)

So what is going on with the dielectric in the capacitor? When you put a charge across the capacitor, the electrons run up the wire and stop at the dielectric. They can't go through, and so they displace the orbits of the electrons inside the dielectric. This act stores energy inside the capacitor.

One commonly used example to explain this and how it works is to talk about hooking a battery up to a capacitor and a light. When the charge builds up in the capacitor and goes to the bulb, the light turns on. However, when the battery is removed from the system, it begins to rely on the energy stored in the capacitor. The energy loops around the circuit, into the capacitor, where it is pulled into the bulb. The energy is used, and more is taken from the capacitor. This slow decline of energy will result in the bulb slowly fading in brightness, rather than simply turning off.

Step 2: Calculations for a Capacitor

$$C = \epsilon_0 \epsilon_r \frac{A}{d}$$

Okay, so the most important calculation here is $C = \epsilon_0 \epsilon_r (A/d)$.

So, what does this mean? Here's a basic translation:

ϵ_0 : A constant, equal to $8.85 * 10^{-12}$ (it's known as the permittivity of free space, and is a constant)

ϵ_r : The dielectric constant (The value assigned to the dielectric based off of what

material it is. Every material has a different dielectric constant, and is always above 1. Resources for dielectric constants can be found on the next step.)

A : This is the area of the plate, recorded in meters²

d: This is the space between the two plates (which is the same as the thickness of the dielectric), and is also recorded in meters²

See the photo for more clarification. The next step is a list of different dielectric constants.

Step 3: Dielectric Constants for Several Common Materials

 ϵ_r

So, here are a few common ones:

Water (room temperature): 80.4

Wax: 2.4-6.5 (depending on what type of wax it is)

Teflon®: 2.0

Air: 1

Lexan®: 2.9

For some less common ones, I have found this

website: <http://www.deltacnt.com/99-00032.htm>

It includes the dielectric constant for a large number of chemicals, as well as other things (Like yeast, for example).

For even more dielectric constants, check out this

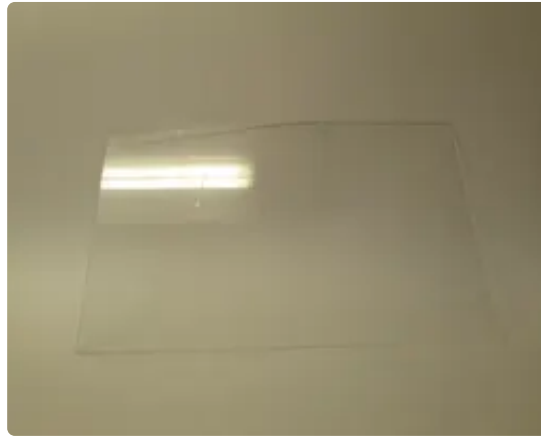
list: <http://www.clippercontrols.com/pages/Dielectric-Constant-Values.html>

It is a much longer and detailed list than the one previously mentioned.

If you can't find the material you are looking for in either of these locations, try a Google search. For example, I wanted to know the dielectric constant of Lexan, so I Googled that, and it turns out the company that manufactures the plastic had

published a file on the specifications of Lexan- and one of those specifications was the Dielectric constant.

Step 4: Making a Basic Parallel Plate Capacitor 1: Picking the Parts



So, as described before, a parallel plate capacitor is a capacitor in which there is two parallel plates separated by a dielectric. In order to build a capacitor, you have to know what materials you have on hand.

I had Lexan and some aluminum tape. They would be easy enough to use, so I picked them.

If you are looking for aluminium tape, try a hardware store. It is used to repair ducts in the heating systems of homes.

Step 5: Making a Basic Parallel Plate Capacitor 2: Assembly



Now for the assembly. This is actually really simple, so i'll just jump right in:

- 1) Take the plate and put it on the underside of the dielectric.
- 2) Secure with adhesive. Make sure that there are no air gaps, etc. (My plate was some aluminium tape, so I just had to make sure that it was on with no air gaps.)
- 3) Place the second plate on the other side of the dielectric. Make sure that it is precisely over where the other plate is- any points that do not overlap are not contributing to the capacitor.
- 4) Secure second plate with adhesive.
- 5) Attach leads to each of the plates.

And with that, your capacitor is ready!

Step 6: Predicting the Outcome



So, in order to predict the outcome of the capacitor I built, we will need to know the dielectric constant of Lexan, the area covered by the plates, and the spacing of the plates.

Here's the formula: $C = \epsilon_0 \epsilon_r (A/d)$

And here's what we know:

ϵ_0 : $8.85 * 10^{-12}$ (because it's a constant)

ϵ_r : 2.9 (because that's the dielectric constant of Lexan)

A : .003521 meters (because the area that the plates overlap is .05257*.06699 in meters, according to Area=Length*Width)

d : .0205 (because that is the spacing of the two plates in meters, which we know because it is the thickness of the Lexan)

So, now for some math:

$$(8.85 * 10^{-12}) * (2.9) * (.003521 / .0205) = 4.4081 * 10^{-12}$$

The answer should be $4.4081 * 10^{-12}$ picofarads, or .0000000000044081 Farads.

Step 7: Proving the Outcome



So, we have now successfully predicted what the outcome will be. Now to test if we are correct:

I started by putting my meter into capacitance mode. When I did this, it read that there was a capacitance of 2.2 picofarads even though there was nothing attached. I wrote this number down so that I could subtract it later- it would be the equivalent of zeroing the meter.

After I had done this, I attached the two ends of the capacitor to the meter; It didn't matter which way. I got a reading of 19.5 picofarads. After subtracting the number I had when there was nothing attached, I found that the capacitance of my capacitor was 17.3 picofarads, or .00000000000173 farads.

As a test to make sure the capacitor was working correctly, I had the meter test resistance. As you can see, it was such a high resistance that the meter was unable to calculate it. This means that there is no short-circuits.

Step 8: Error Calculations

So we calculated .00000000000440 Farads

and we found .00000000000173 Farads when tested.

This is a difference of .00000000000267 Farads.

Unfortunately, the number that we calculated was off slightly from the number found. Why might that be the case?

1) Fringing: In an ideal world, a capacitor would only have an electrical field in between the two panels. However, this is not always the case, and so when the field of the capacitor goes beyond the two fields, engineers call it fringing.

2) Imperfect measuring of dimensions/dimensions not perfect: When measured, we assumed that the plates were perfect rectangles. However, this was not the case, and that may have resulted in the result being thrown off.

3) Improper dielectric constant: I took the dielectric constant of Lexan from their website. However, not all plastics are created equal- even though they are the same type of plastic, and may have been made at the same factory on the same day, they will have a different dielectric constant.

4) Another thing to remember is that the aluminium tape was held on with an adhesive. As user jmwells said, "[The adhesive is} usually a latex base, this could throw your calculations off by 10%"

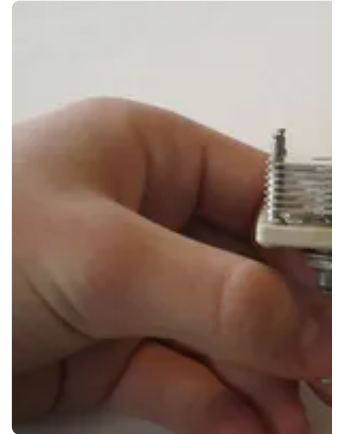
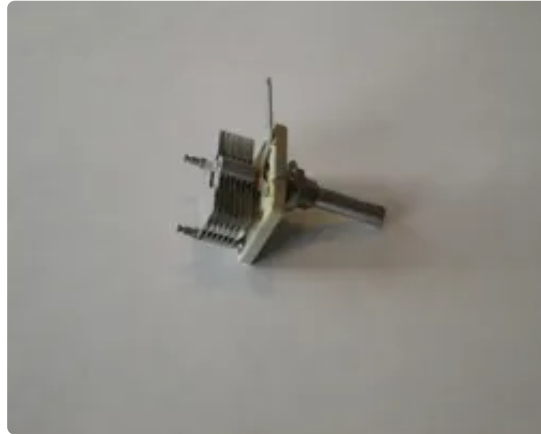
Whatever the case may be in different capacitors, the issue here is clearly fringing. I know this because there is something that I did not take into account when building

the capacitor. There is an engineering rule of thumb rule with these things: If the ratio to the smallest length and the space between the two plates is about less than fifty, then there will be fringing. If it is above, there will still be fringing, but it will be small enough were it can be ignored.

In order to reduce fringing, I should have made sure that the ratio was above fifty. In addition, cutting the shape of the capacitor into a circle would help, as you get less edge and more area that way.

Despite that being the case, I'm pretty pleased with only being off by .00000000000267 Farads, especially when in doing work like this, being off by 10% is considered fantastically well done, unless it is a precision product.

Step 9: So You Want More Capacitance?



If you want to make a capacitor for a hobby project, and you need it to have specific capacitance, odds are you will need more capacitance than a few picofarads. In order to get more capacitance, look at the formula from before:

$$C = \epsilon_0 \epsilon_r (A/d)$$

In order to get a higher capacitance, we can do a few things:

- Make the dielectric constant larger: Pick a new material that will give you a better result.

- Make the area of the plate larger: This can be done easily but is more space consuming.

- Make the space between plates less: This will give you less to divide by, resulting in a larger number, but may be difficult.

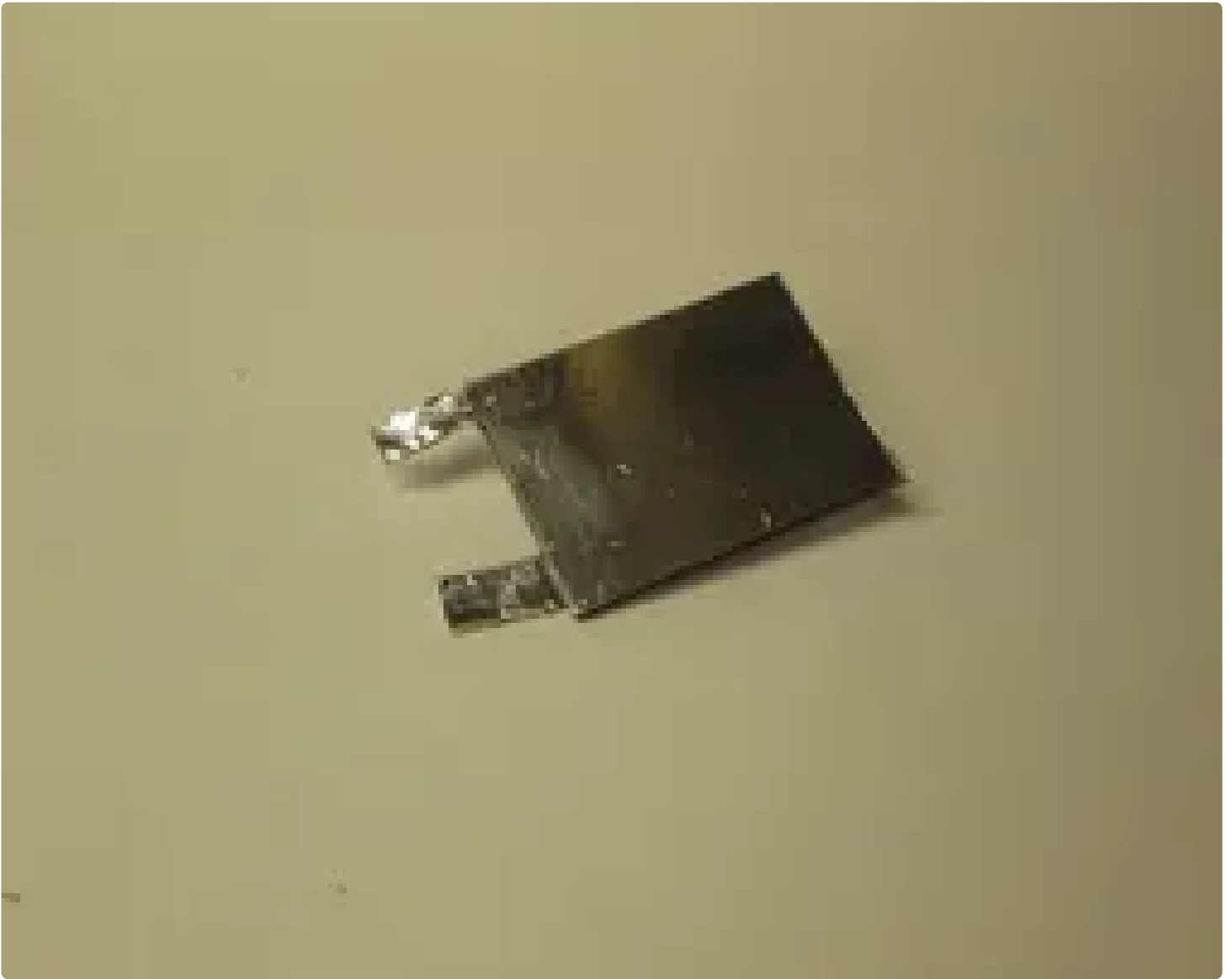
Your final option is to do what they do in industry: Stack a whole bunch of them

together. The first few images are an excellent example of this. It is a variable capacitor that might be used in a radio(they don't all line up perfectly because it is a variable capacitor- by turning the knob, the amount the plates overlap adjust, changing the capacitance.) How is this done? As you can see, there are a large number a plates stacked over -but not touching- each other. In this scenario, air is acting as the dielectric constant, and it generates a similar capacitance to the one I built. There is actually a large number of capacitors there making up one big one, in which the top plate of the capacitor becomes the bottom plate of another.

This can be done with astonishing effectiveness: The last photo is a circular parallel plate capacitor. It has many, many, many layers upon layers, and because there are so many of these thin layers, a very large amount of capacitance can be generated. For example, despite this capacitor only being barely one-twentieth of the size of my capacitor, it gives of more than 1000 times the capacitance.

By doing this, along with the calculation provided, it should be fairly simple to make a functioning capacitor to your specifications.

Step 10: TaDa!



I hope you learned about Parallel Plate Capacitors and how they work. If you build your own, be sure to let me know- I would love to see it! Please feel free to vote for this instructible if you enjoyed it, and as always, have a nice day!

/Assemblyrequired